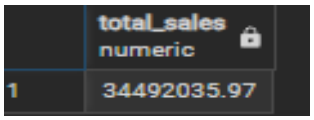
SQL PROJECT – E-COMMERCE ANALYSIS

QUERIES –

KPI'S

1.TOTAL SALES

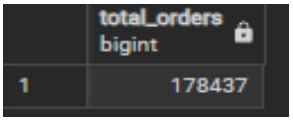
select sum(sales) as total_sales from ecommerce;



	total_sales numeric
1	34492035.97

2. TOTAL ORDERS

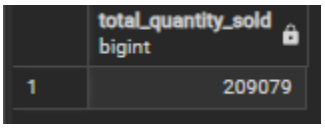
SELECT COUNT (DISTINCT ORDER_ID) AS TOTAL_ORDERS FROM ECOMMERCE;



	total_orders bigint
1	178437

3. TOTAL QUANTITY SOLD

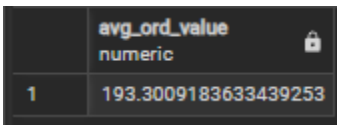
SELECT SUM(QUANTITY_ORDERED) AS TOTAL_QUANTITY_SOLD FROM ECOMMERCE;



	total_quantity_sold bigint
1	209079

4, AVERAGE ORDER VALUE

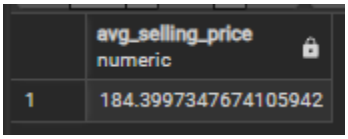
SELECT SUM(SALES) / COUNT(DISTINCT ORDER_ID) AS AVG_ORD_VALUE FROM ECOMMERCE;



	avg_ord_value numeric
1	193.3009183633439253

5 . AVERAGE SELLING PRICE

SELECT AVG(PRICE_EACH) AS AVG_SELLING_PRICE FROM ECOMMERCE;



	avg_selling_price numeric
1	184.3997347674105942

ANALYSIS

6.SALES BY PRODUCT CATEGORY

```
SELECT PRODUCT_CATEGORY,SUM(SALES) AS SALES_BY_CAT FROM ECOMMERCE
GROUP BY PRODUCT_CATEGORY;
```

	product_category character varying (80)	sales_by_cat numeric
1	Audio Devices	3941193.86
2	Batterie	198859.23
3	Charging Cables	633595.40
4	Entertainment Devices	1445700.00
5	Home Appliances	787200.00
6	Laptops and Computers	12167558.70
7	Monitors	6377228.78
8	Phones and Accessori...	8940700.00

7, .TOP 10 PRODUCTS BASED ON TOTAL SALES

```
SELECT PRODUCT,SUM(SALES) AS TOTAL_SALES FROM ECOMMERCE

GROUP BY PRODUCT ORDER BY TOTAL_SALES

DESC LIMIT 10;
```

	product character varying (100)	total_sales numeric
1	Macbook Pro Laptop	8037600.00
2	iPhone	4794300.00
3	ThinkPad Laptop	4129958.70
4	Google Phone	3319200.00
5	27in 4K Gaming Monitor	2435097.56
6	34in Ultrawide Monitor	2355558.01
7	Apple Airpods Headphones	2349150.00
8	Flatscreen TV	1445700.00
9	Bose SoundSport Headphon...	1345565.43
10	27in FHD Monitor	1132424.50

8.TOTAL SALES BY EACH CITY HIGHEST TO LOWEST SALES

```
SELECT CITY,SUM(SALES) AS TOTAL_SALES FROM ECOMMERCE
```

GROUP BY CITY ORDER BY TOTAL_SALES DESC;

	city character varying (50)	total_sales numeric
1	San Francisco	8262203.91
2	Los Angeles	5452570.80
3	New York City	4664317.43
4	Boston	3661642.01
5	Atlanta	2795498.58
6	Dallas	2767975.40
7	Seattle	2747755.48
8	Portland	2320490.61
9	Austin	1819581.75

9.TOTAL SALES BY EACH MONTH

SELECT MONTH,SUM(SALES) AS TOTAL_SALES FROM ECOMMERCE GROUP BY MONTH ORDER BY MONTH ;

	month integer	total_sales numeric
1	12	4613443.34
2	10	3736726.88
3	4	3390670.24
4	11	3199603.20
5	5	3152606.75
6	3	2807100.38
7	7	2647775.76
8	6	2577802.26
9	8	2244467.88
10	2	2202022.42
11	9	2097560.13
12	1	1822256.73

10.TOTAL SALES BY EACH HOUR

SELECT HOUR,SUM(SALES) AS TOTAL_SALES FROM ECOMMERCE GROUP BY HOUR ORDER BY HOUR;

	hour integer	total_sales numeric
1	0	713721.27
2	1	460866.88
3	2	234851.44
4	3	145757.89
5	4	162661.01
6	5	230679.82
7	6	448113.00
8	7	744854.12
9	8	1192348.97
10	9	1639030.58
11	10	1944286.77
12	11	2300610.24
13	12	2316821.34
14	13	2155389.80
15	14	2083672.73
16	15	1941549.60
17	16	1904601.31
18	17	2129361.61
19	18	2219348.30
20	19	2412938.54
21	20	2281716.24
22	21	2042000.86
23	22	1607549.21

11.TOTAL SALES BY EACH TIME OF DAY

SELECT TIME_OF_DAY,SUM(SALES)AS TOTAL_SALES FROM ECOMMERCE GROUP BY TIME_OF_DAY;

	time_of_day character varying (30)	total_sales numeric
1	Afternoon	12531396.39
2	Evening	11742857.59
3	Morning	8269243.68
4	Night	1948538.31

12. TOTAL SALES FOR EACH DAY OF WEEK

SELECT TO_CHAR(ORDER_DATE,'FMDAY')AS DAYNAME,SUM(SALES) AS TOTAL_SALES FROM ECOMMERCE GROUP BY TO_CHAR(ORDER_DATE,'FMDAY');

	dayname text	total_sales numeric
1	Thursday	4839465.16
2	Saturday	4904357.01
3	Tuesday	5087956.78
4	Friday	4855938.38
5	Monday	4883326.72
6	Wednesd...	4988822.26
7	Sunday	4932169.66

13.total quantity sold for each product category and arrange them from highest to lowest

SELECT PRODUCT_CATEGORY,SUM(QUANTITY_ORDERED) AS TOTAL_QUANTITY FROM ECOMMERCE GROUP BY PRODUCT_CATEGORY ORDER BY TOTAL_QUANTITY DESC;

	product_category character varying (80)	total_quantity bigint
1	Batterie	58652
2	Audio Devices	49675
3	Charging Cables	47192
4	Monitors	24122
5	Phones and Accessori...	14449
6	Laptops and Computers	8858
7	Entertainment Devices	4819
8	Home Appliances	1312

14. number of unique orders for each product category and arrange them from highest to lowest

SELECT PRODUCT_CATEGORY,COUNT(DISTINCT ORDER_ID) AS TOTAL_ORDERS FROM ECOMMERCE GROUP BY PRODUCT_CATEGORY ORDER BY TOTAL_ORDERS DESC;

	product_category character varying (80)	total_orders bigint
1	Audio Devices	47446
2	Charging Cables	43359
3	Batterie	41048
4	Monitors	23962
5	Phones and Accessori...	14406
6	Laptops and Computers	8842
7	Entertainment Devices	4794
8	Home Appliances	1312

15 TOP 10 PRODUCTS BASED ON QUANTITY SOLD

SELECT PRODUCT,SUM(QUANTITY_ORDERED) AS TOTAL_ORDERS FROM ECOMMERCE

GROUP BY PRODUCT ORDER BY TOTAL_ORDERS DESC LIMIT 10;

	product character varying (100)	total_orders bigint
1	AAA Batteries (4-pack)	31017
2	AA Batteries (4-pack)	27635
3	USB-C Charging Cable	23975
4	Lightning Charging Cable	23217
5	Wired Headphones	20557
6	Apple Airpods Headphones	15661
7	Bose SoundSport Headphon...	13457
8	27in FHD Monitor	7550
9	iPhone	6849
10	27in 4K Gaming Monitor	6244

AADVANCE ANALYSIS

16. Using CASE WHEN, classify each order line into a sales category:

```
select

case when sales >=1000 then 'high_sales'

        when sales >=500 then 'mediumsales'

           else 'low_sales'
```

```
                        end AS sales_level,
  count(*) as TOTAL_RECORDS  from ecommerce
group by
case when sales >=1000 then 'high_sales'
     when sales >=500 then 'medium_sales'
     else 'low_sales'    end;
```

	sales_level text	total_records bigint
1	high_sales	4740
2	low_sales	163365
3	medium_sal...	17845

17.Using a CTE, calculate the total sales for each product, then find only the products whose total sales are greater than the average sales of all products

```
WITH PRODUCT_SALES AS (
 SELECT PRODUCT,SUM(SALES) AS TOTAL_SALES FROM ECOMMERCE
 GROUP BY PRODUCT)
SELECT * FROM PRODUCT_SALES WHERE TOTAL_SALES >
(SELECT AVG(TOTAL_SALES) AS AVG_SALES FROM PRODUCT_SALES);
```

	product character varying (100)	total_sales numeric
1	27in 4K Gaming Monitor	2435097.56
2	34in Ultrawide Monitor	2355558.01
3	Apple Airpods Headphon...	2349150.00
4	Google Phone	3319200.00
5	iPhone	4794300.00
6	Macbook Pro Laptop	8037600.00
7	ThinkPad Laptop	4129958.70

18. Using a CTE and RANK( ), find the top 3 products by total sales within each product category

```
WITH PRODUCT_SALES AS (

SELECT PRODUCT_CATEGORY,PRODUCT,SUM(SALES) AS TOTAL_SALES,

ROW_NUMBER() OVER(PARTITION BY PRODUCT_CATEGORY ORDER BY
SUM(SALES)DESC) AS RANK FROM ECOMMERCE

GROUP BY PRODUCT_CATEGORY,PRODUCT)

SELECT * FROM PRODUCT_SALES

WHERE RANK <=3
```

	product_category character varying (80)	product character varying (100)	total_sales numeric	rank bigint
1	Audio Devices	Apple Airpods Headphones	2349150.00	1
2	Audio Devices	Bose SoundSport Headphon...	1345565.43	2
3	Audio Devices	Wired Headphones	246478.43	3
4	Batterie	AA Batteries (4-pack)	106118.40	1
5	Batterie	AAA Batteries (4-pack)	92740.83	2
6	Charging Cables	Lightning Charging Cable	347094.15	1
7	Charging Cables	USB-C Charging Cable	286501.25	2
8	Entertainment Devices	Flatscreen TV	1445700.00	1
9	Home Appliances	LG Washing Machine	399600.00	1
10	Home Appliances	LG Dryer	387600.00	2
11	Laptops and Computers	Macbook Pro Laptop	8037600.00	1
12	Laptops and Computers	ThinkPad Laptop	4129958.70	2
13	Monitors	27in 4K Gaming Monitor	2435097.56	1
14	Monitors	34in Ultrawide Monitor	2355558.01	2
15	Monitors	27in FHD Monitor	1132424.50	3
16	Phones and Accessori...	iPhone	4794300.00	1
17	Phones and Accessori...	Google Phone	3319200.00	2
18	Phones and Accessori...	Vareebadd Phone	827200.00	3

19 . For each product category, calculate the total sales, the category's percentage contribution to overall sales, and rank the categories from highest to lowest sales

```
select product_category,sum(sales) as total_sales,
```

rank() over(order by sum(sales) desc) as rk,

round(sum(sales) * 100 / nullif(sum(sum(sales)) over (),0),2) as per_cont from ecommerce

group by product_category order by total_sales desc;

	product_category character varying (80)	total_sales numeric	rk bigint	per_cont numeric
1	Laptops and Computers	12167558.70	1	35.28
2	Phones and Accessori...	8940700.00	2	25.92
3	Monitors	6377228.78	3	18.49
4	Audio Devices	3941193.86	4	11.43
5	Entertainment Devices	1445700.00	5	4.19
6	Home Appliances	787200.00	6	2.28
7	Charging Cables	633595.40	7	1.84
8	Batterie	198859.23	8	0.58

20. Calculate a **running total of monthly sales**

with monthly_sales as (

select extract(month from order_date)as month,

extract(year from order_date) as year,

sum(sales) as monthly_sales

from ecommerce group by extract (month from order_date),extract (year from order_date))

select month,year,monthly_sales,sum(monthly_sales) over (order by month) as running_totals

from monthly_sales order by month,year;

	month numeric	year numeric	monthly_sales numeric	running_totals numeric
1	1	2019	1813586.44	1822256.73
2	1	2020	8670.29	1822256.73
3	2	2019	2202022.42	4024279.15
4	3	2019	2807100.38	6831379.53
5	4	2019	3390670.24	10222049.77
6	5	2019	3152606.75	13374656.52
7	6	2019	2577802.26	15952458.78
8	7	2019	2647775.76	18600234.54
9	8	2019	2244467.88	20844702.42
10	9	2019	2097560.13	22942262.55
11	10	2019	3736726.88	26678989.43
12	11	2019	3199603.20	29878592.63
13	12	2019	4613443.34	34492035.97

21 **Find the top 3 cities by total sales, and calculate each city's percentage contribution to overall sales.**

with cte as

(select city,sum(sales) as total_sales,

row_number() over (order by sum(sales)desc) as rn from ecommerce group by city)

select city,total_sales, round(total_sales * 100/ nullif(sum(total_sales) over (),0),2) as pt_contribution from cte

where rn <=3 order by total_sales desc

	city character varying (50)	total_sales numeric	pt_contribution numeric
1	San Francisco	8262203.91	44.95
2	Los Angeles	5452570.80	29.67
3	New York City	4664317.43	25.38